Vaibhav Kanojia

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Education

Delhi Technological University (DCE) **August 2023 – May 2027**
Bachelor of Technology in Computer Science ***CGPA: 8.5***

ASN Senior Secondary School, Delhi **2023**
Class 12, CBSE ***Percentage: 95.6%***

ASN Senior Secondary School, Delhi **2021**
Class 10, CBSE ***Percentage: 95.5%***

Work Experience

ML Research Intern | IGDTUW **January 2025 – March 2025**

- Architected a **4-stage self-learning CBT chatbot** leveraging BERT-based sentiment analysis, BERTopic-driven topic modelling, and unsupervised clustering to surface cognitive distortions directly from unlabeled conversational data.
- Authored Python preprocessing and analysis prototypes over **∼300 simulated dialogue samples**, formalizing open research gaps in **hallucination control**, persona conditioning, and long-horizon dialogue safety for downstream publication.

AI & Machine Learning Intern | Infosys **October 2024 – December 2024**

- Shipped an **end-to-end OCR pipeline** in **PyTorch + OpenCV**, training a **custom convolutional architecture** on a **60K-sample corpus** with automated preprocessing, normalization, and augmentation to harden the model against real-world noise.
- Stress-tested the network against **LeNet** and **ResNet-18** under identical augmentation, optimizer, and seed protocols; the custom model **beat ResNet-18 by ∼0.4% accuracy** at **one-third the parameter count** and **half the training time**, served behind a **reproducible batch-inference pipeline** at **1K samples per cycle**.

Publications & Research

Divide and Translate: Parameter Isolation with Encoder Freezing for Low-Resource Indic NMT | Link
IJCNLP–AACL 2025

- Proposed a parameter-isolation framework using a **frozen NLLB-600M encoder coupled with 8 direction-specific decoders** to eliminate negative transfer in ultra-low-resource Indic translation while preserving cross-lingual semantic representations.
- Achieved leaderboard scores of **171.4 / 161.1** with **<6% public–private drift** across Hindi↔Bhili/Gondi/Mundari and English↔Santali, evaluated under a weighted BLEU–chrF protocol on the MMLoSo 2025 benchmark.

Projects

GitGalaxy – Explore GitHub repo as navigable 3D semantic galaxy | Link | 3D Knowledge Graph, Semantic Search

- Used **jina-embeddings-v2-base-code** to generate 768-dim representations for every typed symbol extracted by **Tree-sitter**, then projected the resulting manifold into 3D with **UMAP** for spatial exploration in WebGL.
- Tuned a **FAISS index** delivering **sub-20ms queries** over the embedded corpus, backed by a content-addressed cache that loads pre-computed repos instantly; runs fully on-device with no API keys.

Project Synapse – Autonomous agent for last-mile delivery disruptions | Link | Agentic AI, Multi-Step Reasoning

- Designed a custom **ReAct loop** (perceive → plan → execute → reflect → close) over a **typed tool registry**, exposing function signatures to the LLM and orchestrating multi-tool resolution plans for last-mile delivery disruptions.
- Built **structured trace logging** for every plan, tool call, and reflection step, with a **3-tier provider fallback** that degrades gracefully to a deterministic mock – the agent runs end-to-end with no internet, no API keys, and no LangChain.

ProtectHer – Risk-aware urban-safety web application | Link | Multi-Stage Computer Vision, Graph Routing

- Built a **4-stage browser pipeline** that fuses lightweight face and attribute detectors with a custom **VGG16 + Dense-head harassment classifier** trained on 1K paired frames to **88.5% accuracy**, running at **30 FPS** locally.
- Implemented a **Dijkstra-based safest-route picker** that re-ranks candidate routes via a **hotspot-weighted edge model**, alongside a **domain-grounded safety assistant** primed with Indian helpline directories and legal context for emergency-first responses.

Skills

Programming Languages: Python, C, C++
Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, OOP
ML / Systems Stack: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, MySQL
Specializations/Certifications: Intro and Intermediate ML on Kaggle, Machine Learning & Deep Learning (Stanford)

Achievements & Extracurriculars

4× National Hackathon wins: 1st @ Quantum Hackathon – IIT Jodhpur · 1st @ Hack The Universe – Delhi University · 2nd @ TensorQuest – IIEST Shibpur · 3rd @ IMPULSE 2025 – NIT Karnataka.
Top 1% in Amazon ML Challenge 2025 (rank **67 / 20,000+**) and Kaggle Playground Series Oct 2024 (rank **22 / 3,858**).
Top 2% in the Wunder Fund Quant Challenge – rank **79 / 5,000**, within **0.015** of the global leader.
Researcher with **Calibre DTU** · Member of **IEEE**, **GDG DTU**, **AIMS** (AI & ML Society).